

Homework 5

1.8.4 Let $M, N > n$ and suppose that f, g are functions on $\mathbb{R}^n$ that satisfy the estimates $|f(x)| \leq A(1 + |x|)^{-M}$ and $|g(x)| \leq B(1 + |x|)^{-N}$ for all $x \in \mathbb{R}^n$. Prove that for some $C = C(n, N, M) > 0$ we have

$$|(f * g)(x)| \leq CAB(1 + |x|)^{-\min(N, M)} \quad \text{for all } x \in \mathbb{R}^n.$$

Proof.

$$\begin{aligned} |(f * g)(x)| &\leq \left| \int_{\mathbb{R}^n} f(x - y)g(y)dy \right| \leq \int_{\mathbb{R}^n} |f(x - y)||g(y)|dy \\ &\leq \int_{\mathbb{R}^n} \frac{A}{(1 + |x - y|)^M} \cdot \frac{B}{(1 + |y|)^N} dy. \end{aligned}$$

Notice that if $M > N$, then reducing the M to N will be larger than the above integrand. If $N > M$ we can reduce N to equal M and it will also be larger. Therefore we can set $G = \min(M, N)$, and replace M, N with G . Splitting the domain into the ball of radius $|x|/2$ and the rest of $\mathbb{R}^n$.

$$\int_{\mathbb{R}^n} \frac{A}{(1 + |x - y|)^G} \cdot \frac{B}{(1 + |y|)^G} dy = \int_{|y| \leq \frac{|x|}{2}} \frac{A}{(1 + |x - y|)^G} \cdot \frac{B}{(1 + |y|)^G} dy \quad (1)$$

$$+ \int_{|y| > \frac{|x|}{2}} \frac{A}{(1 + |x - y|)^G} \cdot \frac{B}{(1 + |y|)^G} dy. \quad (2)$$

Evaluating (1) and (2) individually:

$$\begin{aligned} (1) &\leq \int_{|y| \leq \frac{|x|}{2}} \frac{A}{(1 + \frac{|x|}{2})^G} \cdot \frac{B}{(1 + |y|)^G} dy = AB(1 + |x|/2)^{-G} \int_{|y| \leq \frac{|x|}{2}} (1 + |y|)^{-G} dy \\ &= C_1(G, n)AB(1 + |x|/2)^{-G} \\ &= C_1AB(1 + \frac{|x|}{2})^{-\min(N, M)}. \end{aligned}$$

$$\begin{aligned} (2) &\leq \int_{|y| > \frac{|x|}{2}} \frac{A}{(1 + |x - y|)^G} \cdot \frac{B}{(1 + \frac{|x|}{2})^G} dy = AB(1 + \frac{|x|}{2})^{-G} \int_{|y| > \frac{|x|}{2}} (1 + |x - y|)^{-G} dy \\ &= C_2AB(1 + \frac{|x|}{2})^{-G}. \end{aligned}$$

We have shown that

$$|(f * g)(x)| \leq (1) + (2) \leq (C_1 + C_2)AB(1 + \frac{|x|}{2})^{-G} = C_3AB(1 + \frac{|x|}{2})^{-\min(M, N)}.$$

$$\begin{aligned} C_3AB(1 + \frac{|x|}{2})^{-\min(M, N)} &= C_3AB(\frac{2 + |x|}{2})^{\min(M, N)} = 2^{\min(M, N)} C_3AB(2 + |x|)^{-\min(M, N)} \\ &\leq 2^{\min(M, N)} C_3AB(1 + |x|)^{-\min(M, N)} \\ &= CAB(1 + |x|)^{-\min(M, N)}. \end{aligned}$$

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